

THERMOCHEMICAL PROPERTIES OF SOME METHANESULFONATES OF TRANSITION METALS

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This work continues the systematic investigation of the thermodynamic properties of methanesulfonic acid's salts. It is devoted to the study of the thermochemical properties of $\text{Mn}(\text{CH}_3\text{SO}_3) \cdot 2\text{H}_2\text{O}$, $\text{Co}(\text{CH}_3\text{SO}_3) \cdot 4\text{H}_2\text{O}$ and $\text{Ni}(\text{CH}_3\text{SO}_3) \cdot 4\text{H}_2\text{O}$. Methanesulfonates of 3d-metals can be used in their electrodeposition and extraction of manganese, cobalt and nickel from spent lithium-ion batteries and alloys containing these elements. Data on their enthalpies formation are necessary for correct modeling of phase equilibria.

Manganese methanesulfonate was obtained via reaction of $\text{MnCO}_3 \cdot \text{Mn}(\text{OH})_2$ and $\text{CH}_3\text{SO}_3\text{H}$ in an aqueous solution. Tetrahydrates of nickel and cobalt methanesulfonates $\text{Co}(\text{CH}_3\text{SO}_3) \cdot 4\text{H}_2\text{O}$ and $\text{Ni}(\text{CH}_3\text{SO}_3) \cdot 4\text{H}_2\text{O}$ were synthesized according to the substitution reaction of barium methanesulfonate in an aqueous solution. The obtained samples were identified by X-ray, ICP-OES and TG analysis.

The thermochemical properties of methanesulfonates were investigated by method of solution calorimetry. The enthalpies of solution of $\text{Mn}(\text{CH}_3\text{SO}_3) \cdot 2\text{H}_2\text{O}$, $\text{Co}(\text{CH}_3\text{SO}_3) \cdot 4\text{H}_2\text{O}$ and $\text{Ni}(\text{CH}_3\text{SO}_3) \cdot 4\text{H}_2\text{O}$ were measured in water at 298.15 K in calorimeter Parr 6755 (Parr Instrument Company). The temperature rise in each run was measured by resistance thermometer (Parr 6772). The thermometric sensitivity was $2 \cdot 10^{-4}$ K. The enthalpies of formation of methanesulfonates at 298.15 K were calculated on the basis of the experimental data and $\text{Mn}^{2+}(\text{aq})$, $\text{Co}^{2+}(\text{aq})$, $\text{Ni}^{2+}(\text{aq})$ and $\text{SO}_3\text{CH}_3^-(\text{aq})$ enthalpies of formation. The results are presented in Table. The enthalpy of formation of methanesulfonate ion in solid state was estimated according to the Mostafa's scheme on the basis of the obtained data for all studied methanesulfonates.

Thermochemical properties of methanesulfonates at 298.15 K

Sample	$\Delta_{\text{sol}}H^{\circ}_{298.15}$, $\text{kJ} \cdot \text{mol}^{-1}$	$\Delta_f H^{\circ}_{298.15}$, $\text{kJ} \cdot \text{mol}^{-1}$
$\text{Mn}(\text{CH}_3\text{SO}_3) \cdot 2\text{H}_2\text{O}$	-24.1 ± 0.5	-2138 ± 9
$\text{Co}(\text{CH}_3\text{SO}_3) \cdot 4\text{H}_2\text{O}$	-6.3 ± 0.4	-2564 ± 9
$\text{Ni}(\text{CH}_3\text{SO}_3) \cdot 4\text{H}_2\text{O}$	-15.6 ± 0.3	-2552 ± 9

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